SCALE TD0

Setup

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1 Introduction

In the next few weeks, you will work on a series of exercises with the goal of learning about how to integrate and use NoSQL data sources in a JAVA project, and how to work with a microservices architecture.

To make sure you are ready then, this document will guide you to setup your environment.

Most of the tools presented here are not mandatory, but are heavily recommended, and using different tools may result in unforeseen problems while programming. During work sessions, you will not receive help if the issues you encounter are caused by a different setup from the recommended one, so take this into account.

2 Setup

2.1 Operating system

Any operating system *should* work, but prefer using Linux if possible.

2.2 IDE

It is recommended to use Jetbrains' IntelliJ IDEA **Ultimate** as your IDE. You can download it on **Jetbrains' website**.

The reason Ultimate is preferred over the free Community Edition is its better framework integration (mainly Quarkus, which we will be using) and additional features like the HTTP Client integration.

As students, you should have access to this software for free, you can apply on **this form**. The IDE is also available with a 30-days trial license.

2.3 JAVA

We will be using JAVA 17 for this project. Newer versions of Java should work just fine, but be careful with third-party libraries, especially code-generation libraries such as Lombok, which may have compatibility problems depending on library versions and java versions.

You need to make sure you have the appropriate JRE and JDK versions installed on your system, with the appropriate PATH and environment variables set.

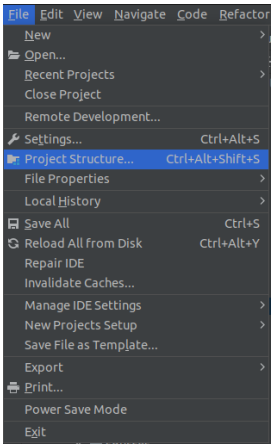

Figure 1 – File menu

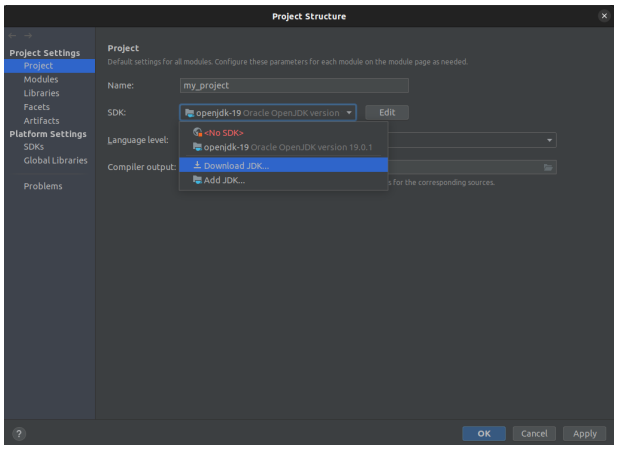

Figure 2 – Project Structure

On debian-based distributions, the two packages you want are `openjdk-17-jdk` and `openjdk-17-jre`. Newer versions will also work.

If you are using IntelliJ IDEA, you can download JAVA directly from the interface, this guarantees that the IDE will be able to find and use your runtime.

To do so :

1. Open any project, then click `File > Project Structure...`
2. In the window that just opened, in the `Project` tab, click the `SDK` selector, then `Download JDK`
3. Select the minimum JAVA version you want (here 17 or above), the vendor (prefer `OpenJDK`), and click `Download`
4. Make sure the `Language level` field is either on “SDK Default”, or at least `17`.

2.4 Maven

Maven will be the build system used in our applications.

IntelliJ IDEA comes bundled with a maven instance. A maven wrapper will also be available in the first session's given files. Should you still decide to use your own installation, use at least Maven 3.8.8.

2.5 Docker

For this course, you will need to be able to run docker containers. Alongside this document, you are given a `docker-compose.yml` file. This is the same file you will be using this week, so check in advance that you are able to launch all of these containers.

The `docker` and `docker-compose` packages should be all you need. The given files were tested using Docker 24.0.7 and Docker Compose 1.29.2.

Important note : avoid running the same service in multiple containers at the same time, as this may lead to unexpected behavior with which image takes priority when accessed from your code. For example, make sure to turn off the MongoDB image used for the NoSQL exercises before starting the MongoDB image used for the development exercises.

2.6 Quarkus

Our framework for these exercises is Quarkus, its robust ecosystem and complete documentation will make development of our services a lot easier. You don't have to prepare anything for it, but it may be a good idea to take a quick glance at its **official documentation** and its **getting started guide**.

If you want to test your environment, you can download a **sample project** which includes some starter code (or simply follow the starter guide). At the root of the project (where the `pom.xml` file is located), you can run the `./mvnw install` command, which will compile your code and generate executable JAR files.

If you want to run your code in development mode and have it automatically refresh as you are coding, you can run the `./mvnw quarkus:dev`.

You may also run your code from the IDEA interface, by opening the `maven` menu on the right side, finding your project, then clicking `Plugins > quarkus > quarkus:dev`. Regular maven commands are accessible in the `Lifecycle` section.

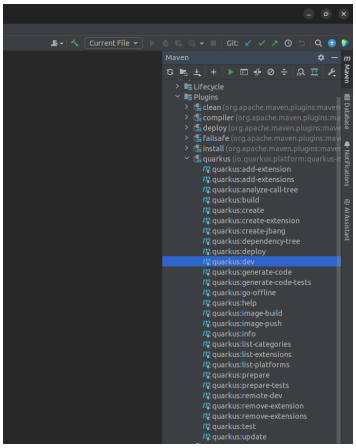

Figure 3 – Maven menu